Answer key: Lesson 3, Digit Cards

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. You have two digit cards: 3 and 5. Make two numbers with both cards.

Answer: 35 and 53. 53 is greater.

With 3 on the left the number is 3 tens and 5 ones: 35. With 5 on the left it is 5 tens and 3 ones: 53. The same two cards make two numbers. 53 is greater, because 5 tens is more than 3 tens.

Watch for: Writing 8 (3 + 5).; Writing 35 twice.

S2. Write the greater number each time.

Answer: 72, 48, 65, 30

(a) 7 tens is more than 2 tens: 72. (b) Same tens, 8 ones is more than 1 one: 48. (c) 6 tens is more than 5 tens: 65. (d) 3 tens is more than 1 ten: 30.

Watch for: Choosing the number with the greater ones digit (saying 13 is greater than 30).; Not looking at the tens first.

S3. Count the bars and squares in Mei's picture. Ben has 4 bars and 1 square.

Answer: Mei 38, Ben 41, Ben has more

Mei has 3 bars and 8 squares: 38. Ben has 4 bars and 1 square: 41. 4 tens is more than 3 tens, so Ben has the greater number, even though Mei has more squares.

Watch for: Saying Mei has more because she has 8 squares.; Reading Mei's picture as 83.

S4. Make two numbers with both cards 6 and 2.

Answer: 62 and 26

The greater card on the left makes the greater number: 62. The smaller card on the left makes the smaller number: 26.

Watch for: Writing 62 for both.; Writing 8 (6 + 2).

S5. Arrange these in order. Begin with the smallest number: 52, 25, 41, 14.

Answer: 14, 25, 41, 52

14 has 1 ten, 25 has 2 tens, 41 has 4 tens, 52 has 5 tens. Smallest first: 14, 25, 41, 52.

Watch for: Putting 25 after 41 because 5 is greater than 1.; Writing them greatest first.

S6. Kumar says that 19 is greater than 21.

Answer: No. 21 is greater. 2 tens is more than 1 ten.

19 is 1 ten and 9 ones. 21 is 2 tens and 1 one. 2 tens is more than 1 ten, so 21 is greater. Kumar looked at the ones digits first. Look at the tens first.

Watch for: Saying Kumar is right because 9 is more than 2.; Not giving a reason (any clear sentence about the tens is fine).

S7. You have the cards 2 and 8. Use both cards each time.

Answer: 82 and 28

82 has 8 tens, which is more than 5 tens. 28 has only 2 tens, which is less than 5 tens.

Watch for: Writing 82 for both parts.; Writing 8 and 2.

S8. Siti has 25 stickers. Ravi has 52 stickers. Draw bars and squares in the box.

Answer: Ravi (52 is 5 tens and 2 ones; 25 is 2 tens and 5 ones)

Siti: 2 bars and 5 squares. Ravi: 5 bars and 2 squares. 5 bars is more than 2 bars, so Ravi has more stickers, even though both numbers use a 2 and a 5.

Watch for: Saying they have the same because they use the same digits.; Drawing 2 and 5 single squares for Siti.

Think harder (page 3)

T1. You have the cards 3, 5 and 8. Use two different cards to make the greatest number. Then make the smallest number.

Answer: 85 and 35

Numbers you can make: 35, 38, 53, 58, 83, 85. The greatest is 85 (8 tens, the greatest card on the left, then the next greatest card). The smallest is 35.

Watch for: Using all three cards.; Writing 88 (using the 8 twice).

T2. You have the cards 0, 3 and 7. Use two cards to make the smallest and the greatest two-digit numbers.

Answer: 30 and 73

The 0 can not go on the left (03 is just 3). The two-digit numbers are 30, 37, 70 and 73. The smallest is 30 and the greatest is 73.

Watch for: Writing 03 or 3 for the smallest.; Writing 37 for the smallest.

T3. You have the cards 1, 3, 5 and 7. Make two-digit numbers with two different cards. The number is more than 40. Its left digit is greater than its right digit. Write all the numbers that fit.

Answer: 51, 53, 71, 73, 75

More than 40 means a tens card of 5 or 7. With 5 on the left: 51, 53, 57. With 7 on the left: 71, 73, 75. Clue 2 removes 57, because 5 is not greater than 7. Left: 51, 53, 71, 73, 75.

Watch for: Stopping after one or two numbers.; Writing 57 (5 is less than 7).

T4. You have the cards 2, 4 and 6. Use two different cards each time. Write all the two-digit numbers you can make.

Answer: 6 numbers: 24, 26, 42, 46, 62, 64

2 on the left: 24, 26. 4 on the left: 42, 46. 6 on the left: 62, 64. That is 2 + 2 + 2 = 6 numbers.

Watch for: Missing some by listing in a random order.; Writing 22, 44 or 66 (each card can be used only once).

Go further (page 4)

G1. You have the cards 2, 5, 6 and 9. Use two different cards to make a two-digit number.

Answer: 59 and 52

The numbers are 25, 26, 29, 52, 56, 59, 62, 65, 69, 92, 95, 96. Less than 60: 25, 26, 29, 52, 56, 59, so the greatest is 59. Greater than 50: 52, 56, 59, 62, 65, 69, 92, 95, 96, so the smallest is 52.

Watch for: Writing 96 or 69 for part a (they are greater than 60).; Writing 25 for part b (it is not greater than 50).

G2. You have the cards 1, 2, 3 and 4. Make two-digit numbers. How many numbers can you make?

Answer: 12, 16 and 6

Each card once: 4 choices for the tens card, then 3 choices for the ones card, so 4 x 3 = 12. A card may be used twice: 4 choices and 4 choices, so 16 (the 12 and the four numbers 11, 22, 33, 44). Part c is beyond what school teaches: 3 cards in a row. 1 first: 123, 132. 2 first: 213, 231. 3 first: 312, 321. That is 3 x 2 x 1 = 6 numbers.

Watch for: Counting only 8 or 6 numbers.; Forgetting 11, 22, 33 and 44 in part b.; Part c: listing in a random order and missing one (there are 6, two for each first card).

G3. I make a two-digit number with two different cards from 1, 3, 4 and 7. My number is between 35 and 45. My ones digit is greater than my tens digit. What number am I?

Answer: 37

The numbers are 13, 14, 17, 31, 34, 37, 41, 43, 47, 71, 73, 74. Between 35 and 45: 37, 41, 43. Ones digit greater than tens digit: 37 (7 is greater than 3). 41 and 43 have a ones digit that is smaller.

Watch for: Stopping at 41 or 43 and not checking the ones digit.; Writing 34 or 47 (they are not between 35 and 45).

Investigation (page 5): Is the greater card always first?

For two different cards, the greater card on the left always makes the greater number, because the tens decide. With the cards 0 and 8 you can only make 80, because 08 is just 8. With the same card twice (4 and 4) you can only make 44. So you can make two different two-digit numbers only when the two cards are different and neither is 0.

Look for: A child who tests several pairs, notices that the tens digit decides, and finds the two special cases (a 0 card, two matching cards). A reason using tens and ones is better than "because it is greater".

Exit ticket (page 5)

E1. Write the greater number: 39 or 41.

Answer: 41

41 has 4 tens and 39 has only 3 tens, so 41 is greater. Look at the tens first.

Watch for: Writing 39 because 9 is greater than 1.

E2. Make two numbers with both cards 7 and 3.

Answer: 73 and 37

The greater card on the left makes the greater number: 73. The smaller card on the left makes the smaller number: 37.

Watch for: Writing 73 for both.; Writing 10 (7 + 3).

E3. Arrange these in order. Begin with the smallest number: 63, 36, 60.

Answer: 36, 60, 63

36 has 3 tens. 60 and 63 both have 6 tens. 60 has 0 ones and 63 has 3 ones, so 60 comes before 63.

Watch for: Putting 60 before 36 or after 63.; Writing them greatest first.

Sneak-peek puzzles (not in the workbook)

1. 18
2. Mei. Her number is 84, which is 8 tens. Ali has 48, which is only 4 tens.